

Kashf

Root-Cause Diagnosis & Scenario Testing for Dubai's Road Network

Business & Implementation Plan — Challenge Deliverable

RTA × CUD 4IR Mobility Student Challenge — Emerging Innovative Technologies Section (TSG)

Working prototype: sudofixit.github.io/kashf- · July 2026 · Version 1.0

Our Understanding of Your Needs

Move congestion management from reactive to proactive — on infrastructure RTA already owns



Find Problems Before Complaints

Systematically identify which corridors and junctions are failing — ranked by real measured shortfall, not by which complaint arrived first.



Explain Why, Not Just Where

Distinguish a signal-timing fault from raw demand overload from weather impairment — because each demands a different response.



Test Fixes Before Spending

Estimate and rank candidate interventions — retiming, capacity, demand management — before budget or political capital is committed.



Fit Existing RTA Systems

Complement — not duplicate — EC3, UTC-UX Fusion and the Databricks data platform. Human approval on every action, full audit trail.

The Gap We Fill

RTA already senses and predicts world-class. The diagnose → test → recommend step is still manual.



Today this fusion happens manually, across separate dashboards, in an operator's head, under time pressure.

Solution Overview — Five Stages

One engine, five plain-English steps, every number traceable to source data

1 • Triage

Scan every monitored corridor and rank the real bottlenecks: how many vehicles want the road versus how many get through, and how often it hits gridlock.

2 • Diagnose

For the worst locations, determine the cause — a failing signal, demand beyond any signal's reach, weather impairment, or plain capacity shortfall.

3 • Simulate

Test candidate fixes with a transparent capacity-delay model. Results are labelled indicative — decision-support, not a substitute for full micro-simulation.

4 • Rank

Score each fix on benefit, cost and disruption, and recommend the best value — with the trade-offs stated, not hidden.

5 • Learn

Log every diagnosis, recommendation and outcome, building an institutional case library that improves future recommendations.

Operator Console

A live map console with plain-English findings, an approvable recommendation card, and "Ask Kashf" — a built-in AI assistant that explains any decision on demand.

How We Use Mistral AI

A strict division of labour: AI reasons and explains — Python owns every number

Mistral does the reasoning

Diagnosis narratives, plain-language explanations of every metric, the recommendation text, and the live "Ask Kashf" operator chat — grounded, in each call, in that case's verified data only.

Python does the arithmetic

Triage rankings, capacity-delay calculations, confidence scores — all computed deterministically. The AI is never asked to produce a number.

The faithfulness gate

Every AI-written sentence is machine-checked against the computed values before it is shown. Any output whose numbers do not exactly match is rejected and regenerated. Plain English can never drift from the real math.

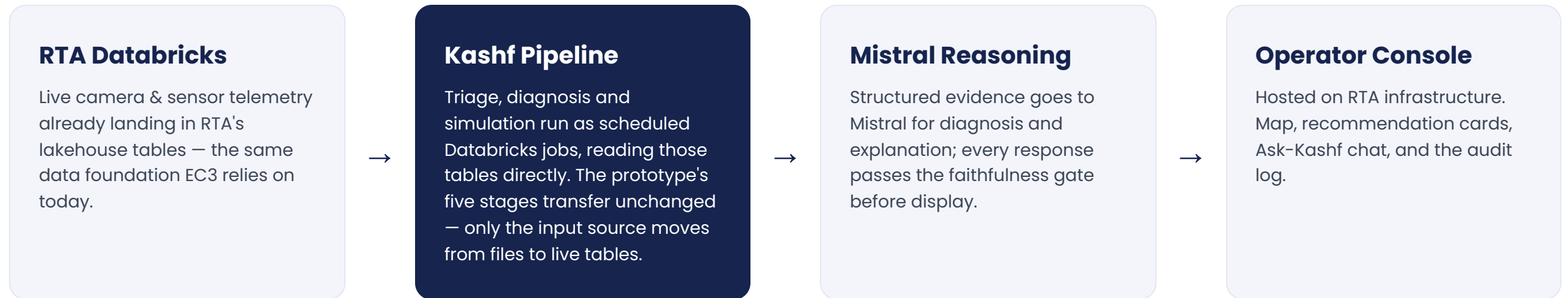
Deployment options

Mistral's EU-hosted API (La Plateforme) for the pilot; Mistral's open-weight models can be self-hosted inside RTA infrastructure if full data residency is required at scale.

Judged criterion, answered directly: Mistral does real work in Kashf — diagnosis, explanation, and live operator Q&A — not decoration.

Integration with RTA's Data Platform

Kashf consumes the Databricks streams RTA already operates — no new data infrastructure



- **Prototype today:** runs on the challenge dataset (26,304 hours × 18 corridors, 2023–2025) — fully working, publicly demonstrable.
- **Production path:** the same pipeline pointed at RTA's live Databricks tables — validated in shadow mode against real operations before any operator relies on it.

AI Governance & Responsible AI

Not a compliance appendix — these controls are already running in the prototype

Human-in-the-loop

Kashf recommends; a person approves, modifies or rejects. Nothing reaches signals, patrols or public messaging without a logged human decision.

Traceable numbers

Every figure on screen traces to source data. AI text is machine-verified against computed values before display — enforced, not aspirational.

Honest confidence

Confidence scores are computed from how many independent signals agree — never self-reported by the AI. Low confidence explicitly requests human review.

Stated limits

Simulation outputs are labelled indicative in the interface itself. The system declines questions outside its data — it does not improvise routes or causes.

Audit trail

Every recommendation, decision, operator identity and timestamp is written to an immutable log — reviewable, attributable, permanent.

No personal data

Kashf uses aggregate traffic measurements only — no plate numbers, no individual journeys, no personal information of any kind.

Implementation Methodology

A phased, evidence-gated delivery — RTA engaged at every stage



Weekly status reviews · shared project plan · single point of contact throughout

Timeline — Six Months to Live Pilot

Evidence gates between phases: no stage advances until the previous one has proven itself

Phase	Months	Scope	Exit gate
Discover & Design	1 – 2	Data landscape & Databricks mapping · EC3 workflow workshops · integration blueprint · thresholds recalibrated on RTA's own data	Signed-off blueprint & governance plan
Build	3 – 4	Pipeline as Databricks jobs · operator console on RTA hosting · Mistral reasoning layer with faithfulness gate · audit logging	End-to-end run on live tables
Shadow Mode & UAT	5	Kashf runs beside real operations without acting · recommendations compared to what operators actually did · accuracy report	Backtest accuracy accepted by RTA
Pilot Go-Live	6	5–10 pilot corridors live · operator training · hypercare · handover documentation	Pilot acceptance & Phase-2 decision

Phase 2 (full corridor network, deeper UTC-UX Fusion integration, learning loop at scale) scoped after pilot results — deliberately not pre-sold.

Cost of Adoption — What Kashf Actually Adds

Kashf is designed to run on what RTA already owns — the incremental spend is minimal, and stated line by line

Line	Incremental cost (AED)	Why
Mistral AI inference	≈ 2,000 – 4,000 / yr	The only genuinely new line item. Current flagship API pricing (\$0.50 in / \$1.50 out per 1M tokens); pilot-scale diagnosis + operator chat totals tens of millions of tokens per year — a few thousand dirhams, not a cost driver. Self-hosting Mistral's open-weight models inside RTA infrastructure is an option if full data residency is preferred.
Data platform & compute	0	The pipeline runs as jobs inside RTA's existing Databricks capacity — no new platform, no new licences.
Hosting	≈ 0 – 15,000 / yr	One application VM + small database on RTA's existing cloud/on-prem estate — marginal capacity, not new procurement.
Map layer	0	The prototype uses Mapbox (free tier covers an internal console). In production, the console renders equally on RTA's own enterprise GIS/mapping stack — the map is a display layer, not a dependency.
Integration effort	Internal allocation	~6 months of RTA's in-house engineering and data team time (2 engineers + part-time data scientist & lead) — effort RTA already staffs for, not money leaving the building. A delivery partner is an option, not a requirement.

Recommended Operating Model

How Kashf stays healthy after go-live — owned and run by RTA teams

Functional & Technical

Configuration fixes, interface monitoring, data-pipeline reconciliation, threshold tuning as traffic patterns evolve.

Engineering

Bug fixes, patches, version upgrades and enhancement requests under defined SLAs.

Model Stewardship

Quarterly re-validation of detection thresholds and the diagnosis layer against the latest data — accuracy is maintained, not assumed.

Hypercare Window

Focused support in the first weeks after go-live — daily check-ins, rapid fixes — before settling into normal operations with a named owner.

Why This Is Credible

Every claim below is verifiable today — nothing is a promise

2.43 kph

Mean forecast error on data the model had never seen — honest, leakage-checked validation

26,304 hrs

of real measurements analysed across 18 corridors and 10 junctions, 2023–2025

0

unverified numbers — every figure in every screen traces to a validated source file

Live

working demo, public Git link — map, diagnosis cards, and Ask-Kashf assistant

- Our own QA caught and corrected errors before they reached this document — including an inflated early accuracy figure and two conflicting statistics. That same discipline is what RTA gets in delivery.
- The prototype's five stages, governance controls and faithfulness gate are running now — the pilot productionises what already works, it does not start from a slide.